

CALIBER 2026

Hackathon Submission

Industry Theme: Healthcare

**Submission Theme: AI-Powered Clinical Operations
Copilot**

Team: Caliber Collective

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Introduction of the Team

Meet the Team – Caliber Collective

We are a cross-functional team with expertise in AI, Azure Cloud, Microsoft Copilot, and healthcare workflows, focused on building secure and scalable enterprise AI solutions.

What Sets Us Apart

- ✓ Copilot Studio & Azure OpenAI expertise
- ✓ Cloud-native & serverless architecture skills
- ✓ Clinical operations & healthcare domain knowledge
- ✓ Focus on security, compliance & responsible AI

Our Mission

Reduce clinician administrative burden and improve care quality through secure, enterprise-grade AI.



Project Overview

Executive Summary

AI-Powered Clinical Operations Copilot is an enterprise-grade, Copilot-based healthcare assistant designed to reduce clinician workload by summarizing patient data, automating routine operational tasks, and supporting efficient care coordination using Microsoft's secure AI ecosystem.

Problem Statement

Clinicians spend a significant portion of their time on manual, non-clinical activities, such as:

- Reviewing fragmented patient records
- Preparing clinical summaries
- Coordinating follow-ups and tasks

This leads to burnout, inefficiencies, and delayed patient care.

Objective

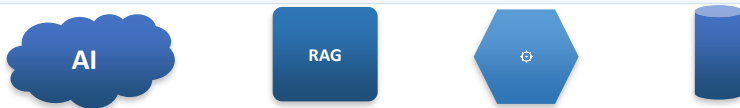
- ✓ Reduce administrative workload for clinicians
- ✓ Enable faster access to patient insights
- ✓ Improve operational efficiency in clinical workflows
- ✓ Ensure data security and healthcare compliance

Technical Architecture

Secure, cloud-native, Retrieval-Augmented Generation (RAG) architecture built entirely on Microsoft Azure & Copilot ecosystem

High-Level Flow

- 1 Clinician interacts with Copilot Studio using natural language
- 2 Copilot sends requests to Azure OpenAI Service
- 3 Relevant patient & operational data retrieved via Azure AI Search
- 4 Business logic & processing handled via Azure Functions
- 5 Clinical workflows orchestrated using Azure Logic Apps
- 6 Data stored in Microsoft Dataverse & Azure Blob Storage



Copilot → Azure OpenAI → AI Search (RAG) → Functions → Storage

Core Architecture Components

Copilot Studio

Conversational interface for clinicians to query summaries & workflows

Azure OpenAI Service

Generates accurate, contextual clinical summaries via enterprise LLMs

Azure AI Search (RAG)

Retrieves patient records securely without exposing raw data

Azure Functions

Serverless backend logic, validations & response orchestration

Azure Logic Apps

Automates follow-ups, task creation & care coordination

Dataverse & Blob Storage

Secure structured & unstructured data with enterprise access control

Technology Justification

- Microsoft-first architecture ensures enterprise security and compliance
- No customer data used for model training
- Scalable and serverless — ideal for healthcare workloads
- Cost-optimized, suitable for lab credits (~\$200)
- Ready for future EHR / FHIR integrations



Enterprise Security

HIPAA Compliant

Zero Trust Architecture

Architecture designed for security, scalability, and responsible healthcare AI.

Solution Architecture

Secure, Cloud-Native RAG Architecture on Microsoft Azure & Copilot Ecosystem

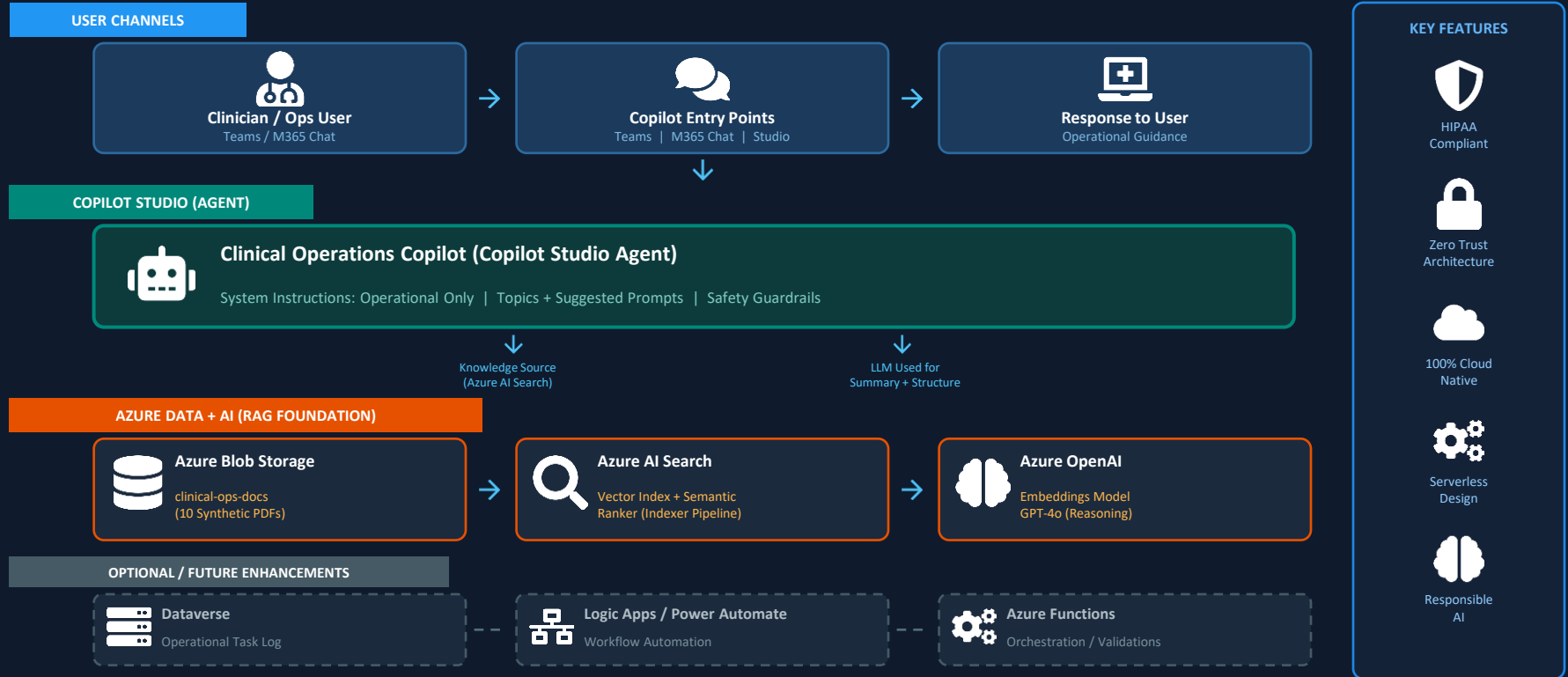


Diagram shows current build + optional enhancements

Architecture designed for security, scalability, and responsible healthcare AI.

Execution Steps

Setup | Execution Flow | Benefits

Setup & Configuration



Copilot Studio Setup

Create agent for clinical ops, define topics & intents



Azure OpenAI Integration

Connect with Azure OpenAI for clinical summarization



Data Preparation

Dataverse + Blob Storage + AI Search indexing



Backend Services

Azure Functions for processing, Logic Apps for workflows

Execution Flow

- 1 Clinician submits query via Copilot
- 2 Copilot retrieves data via AI Search (RAG)
- 3 Azure OpenAI generates clinical response
- 4 Azure Functions process logic & validations
- 5 Logic Apps trigger operational workflows
- 6 Results presented securely to clinician

Operational Benefits

- ✓ Faster access to patient insights
- ✓ Reduced manual administrative effort
- ✓ Automated task & workflow management
- ✓ Secure, compliant & scalable execution

VM Details & Execution Instructions



VM Details

Not Applicable

This solution uses a fully cloud-native, serverless architecture leveraging Microsoft Azure managed services. No Virtual Machines are required.



Execution Environment



Entirely hosted on Microsoft Azure



Uses lab credits (~\$200) for hackathon



Scalable, secure & cost-optimized



No local setup required



How to Execute the Solution

1

Launch Copilot Studio Agent

Configured for Clinical Operations

2

User Interacts via Natural Language

e.g. "Summarize patient details and suggest follow-up actions"

3

Copilot Processes the Request

Azure OpenAI for response generation
Azure AI Search for patient context retrieval

4

Backend Logic & Validations

Azure Functions handle processing & validations

5

Workflow Orchestration

Azure Logic Apps trigger tasks, reminders & coordination

6

Results Displayed Securely

Responses returned to clinician in Copilot interface

✓ *Serverless execution ensures high scalability, low cost, and enterprise-grade security.*

Demo and Results

Live interaction with AI-Powered Clinical Operations Copilot built using Microsoft Copilot Studio and Azure AI Services

✓ Live Demo Access – Clinical Operations Copilot

(Microsoft Copilot Chat & Microsoft Teams)

🔗 **Agent Access Link:** [🔗 Click Here:](#)

📹 **Demo Video Recording:** [🔗 Click Here:](#)

Demo Flow

- 1 Clinician interacts with Copilot using natural language
- 2 Requests patient summaries and operational assistance
- 3 Secure retrieval of patient context via Azure AI Search
- 4 AI-generated, structured clinical insights
- 5 Automated workflow suggestions (follow-ups, tasks, next steps)

Key Scenarios Demonstrated



Patient Summary Generation

Quick, contextual summary of patient history and recent visits



Operational Task Automation

Automated follow-ups and task recommendations via Logic Apps



Care Coordination Support

Smooth hand-offs and next-step guidance across care teams

Results & Impact

✓ Reduced clinician administrative workload

✓ Faster access to actionable patient insights

✓ Improved clinical workflow efficiency

✓ Secure, compliant data handling

Copilot enables clinicians to focus more on patient care and less on administration

Additional Details in Brief



Use Case / Idea

The **AI-Powered Clinical Operations Copilot** assists clinicians by summarizing patient information, automating routine administrative tasks, and supporting care coordination—reducing time spent on non-clinical activities and improving overall care delivery.



Introduction to Solution

A Copilot-based healthcare agent built on Microsoft Azure that enables clinicians to interact using natural language and receive accurate, contextual, and actionable clinical insights securely within an enterprise environment.



Problem Statement

Clinicians spend a significant amount of time on administrative tasks, leading to inefficiencies, burnout, and reduced patient interaction time.



Objectives

- ✓ Minimize clinician administrative workload
- ✓ Improve operational efficiency in healthcare workflows
- ✓ Enhance care coordination using AI
- ✓ Maintain enterprise security and compliance standards

Designed to be secure, scalable, and clinically impactful using Microsoft Copilot and Azure AI.

Capabilities & Architecture

What It Does



Accepts clinician queries through Copilot Studio



Retrieves relevant patient and operational data using Azure AI Search



Generates structured clinical summaries via Azure OpenAI Service



Automates operational workflows (follow-ups, reminders, task creation)



Returns secure, concise responses to clinicians in real time

Architecture (RAG Approach)

The architecture follows a Retrieval-Augmented Generation (RAG) approach:



Patient data stored securely in Dataverse and Azure Blob Storage



Azure AI Search retrieves only relevant context



Azure OpenAI processes prompts without training on customer data



Azure Functions handle backend logic and validations



Azure Logic Apps orchestrate care coordination workflows

Ensures security, scalability, and compliance

Technical Stack & Assumptions

Why These Components



Copilot Studio

Rapid, low-code enterprise conversational AI



Azure OpenAI Service

Secure, private large language models



Azure AI Search

Accurate and compliant context retrieval



Azure Functions

Serverless, scalable backend processing



Azure Logic Apps

Workflow automation for healthcare operations



Dataverse & Blob Storage

Secure structured and unstructured data storage



Assumptions



Synthetic or anonymized patient data used for demonstration



Users authenticated via enterprise identity



Solution implemented as a PoC for hackathon purposes



Designed to be extensible for real EHR integrations

Future Roadmap & Next Steps

Strategic Evolution Plan for AI-Powered Clinical Operations Copilot



Thank You!

Team: Caliber Collective

AI-Powered Clinical Operations Copilot

Thank you for the opportunity to present our solution.

Our Goal

Empower clinicians by reducing administrative burden and enabling better patient care through secure, enterprise-grade AI.

Built using Microsoft Copilot and Azure AI services, our solution demonstrates how responsible AI can transform clinical operations at scale.

Let clinicians focus on care — while Copilot takes care of the rest.



OUR PILLARS

Transforming healthcare through secure, scalable, and responsible AI innovation.